MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, DIGITAL, KEY CODE SBT

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface receiver module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.057 failures per million operating hours for class I or 0.086 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SBT.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/23-1 for class I and M28787/23-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

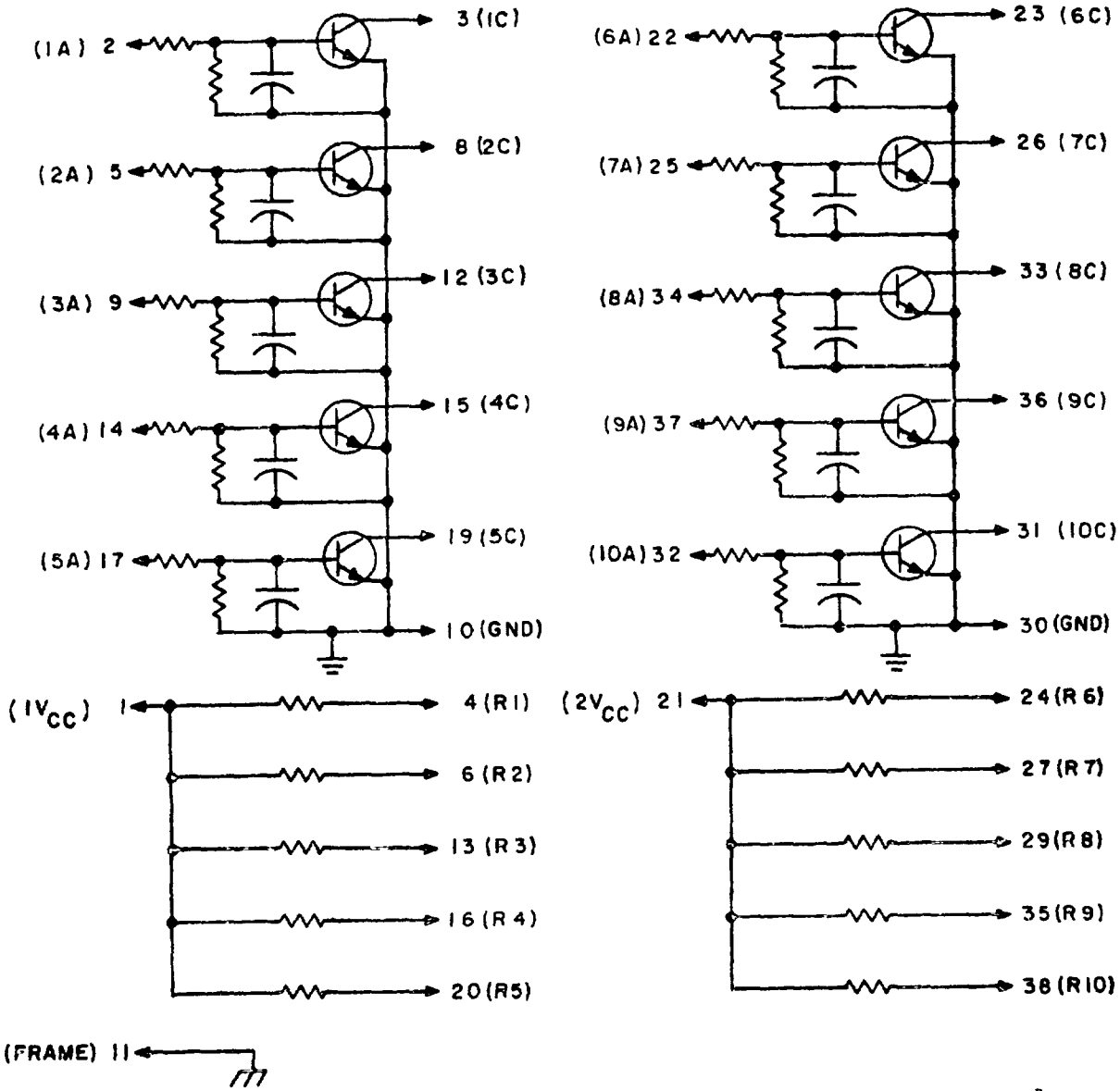
3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

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NOTES:
1. A = Input, C = Open collector.
2. R1-R10 = 4.64 kΩ ±1%, 1/20 W.
3 The following pins are unused and each shall be isolated from all other pins: 7, 18, 28, 39, and 40.

* FIGURE 1. SBT module function diagram, interface receiver.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25° ±5°C 3/		Class I 3/				Class II 3/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A10	Breakdown voltage check	II	S	0	5	0	5	0	5	0	5	0	5	µA
B1-B10	Output voltage, low level (V_{IN} = 22 V, I_C = 50 mA)													
	Initial	III	X	0	250	0	350	0	350	0	350	0	350	mV
	End-of-life	III	X	0	300	0	400	0	400	0	400	0	400	mV
C1-C10	Output voltage, high level (V_{IN} = Open)													
	Initial	IV	X	4.90	5.05	4.85	5.05	4.85	5.05	4.85	5.05	4.85	5.05	V
	End-of-life	IV	X	4.85	5.05	4.40	5.05	4.40	5.05	4.40	5.05	4.40	5.05	V
D1-D10	Output voltage, high level (V_{IN} = 6 V)													
	Initial	V	S	4.90	5.05	4.85	5.05	4.85	5.05	4.85	5.05	4.85	5.05	V
	End-of-life	V	S	4.85	5.05	4.40	5.05	4.40	5.05	4.40	5.05	4.40	5.05	V
E1-E10	Input current, low level (V_{IN} = 6 V)	VI	S	0	285	0	285	0	285	0	285	0	285	µA
F1-F10	Input current, high level (V_{IN} = 22 V)	VII	X	0.10	1.15	0.10	1.15	0.10	1.15	0.10	1.15	0.10	1.15	mA
G1-G10	Input pulse noise suppression													
	Initial	VIII	X	4.75	5.05	4.70	5.05	4.70	5.05	4.70	5.05	4.70	5.05	V
	End-of-life	VIII	X	4.70	5.05	4.70	5.05	4.70	5.05	4.70	5.05	4.70	5.05	V
H1-H10	Propagation delay time (t_{PHL})													
	Initial	IX	X	250	450	250	600	250	600	250	600	250	600	µs
	End-of-life	IX	X	250	600	243	708	243	708	243	708	243	708	µs

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25° ±5°C 3/		Class I 3/				Class II 3/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J10	Propagation delay time (t_{PLH})													
	Initial	X	X	96	172	75	230	75	230	75	230	75	230	µs
	End-of-life	X	X	96	230	82	271	82	271	82	271	82	271	µs
K1-K6	Isolation													
	Initial	XI	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XI	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
L1-L10	Resistance													
	Initial	XII	X	4.588	4.692	4.506	4.774	4.506	4.774	4.506	4.774	4.506	4.774	kΩ
	End-of-life	XII	X	4.506	4.774	4.423	4.857	4.423	4.857	4.423	4.857	4.423	4.857	kΩ
M	Continuity	XIII	S	0	1	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ Unless otherwise specified, limits are for initial and end-of-life.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through M shall be performed in any order with the following exceptions. If test conditions A or K are part of a sequence of tests, the order of testing shall be as follows: A, K where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through M in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through L shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the oerating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through M shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Cross reference of classification. Military Specification MIL-M-28787/23 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658676. Part number M28787/23-1 supersedes part numbers 2658676 and M28787/23.

* 6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)			
Output, low level	0.2	0.2	W
Output, high level	1.0	1.0	W
Input voltage			
Low level	6.0 max	6.0 max	V dc
High level	22.0 min, 32.0 max	22.0 min, 32.0 max	V dc
Input current			
Low level (V_{IL} = 6 V)	0 min, 285 max	0 min, 285 max	µA
High level (V_{IH} = 22 V)	0.1 min, 1.15 max	0.1 min, 1.15 max	mA
Output conditions			
Collector-to-emitter breakdown voltage	50 min	50 min	V
Transistor-on voltage (I_C = 50 mA, V_{IH} = 22 V)	0 min, 0.4 max	0 min, 0.4 max	V
Transistor-off leakage current (V_{CE} = 50 V)	0 min, 5 max	0 min, 5 max	µA
Propagation delay time (14 mA load)			
t_{PHL}	243 min, 708 max	243 min, 708 max	µs
t_{PLH}	82 min, 271 max	82 min, 271 max	µs

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* 6.6 Module function. This module contains 10 identical interface receiver circuits which can be used to convert +28 volt logic levels to 0 and 5 V logic levels. This module also contains 10 resistors for use as current limiting resistors. Each circuit incorporates low-pass filtering at the input to prevent noise transients from affecting the output logic levels.

6.6.1 Parallel outputs. Paralleling of outputs is not permissible.

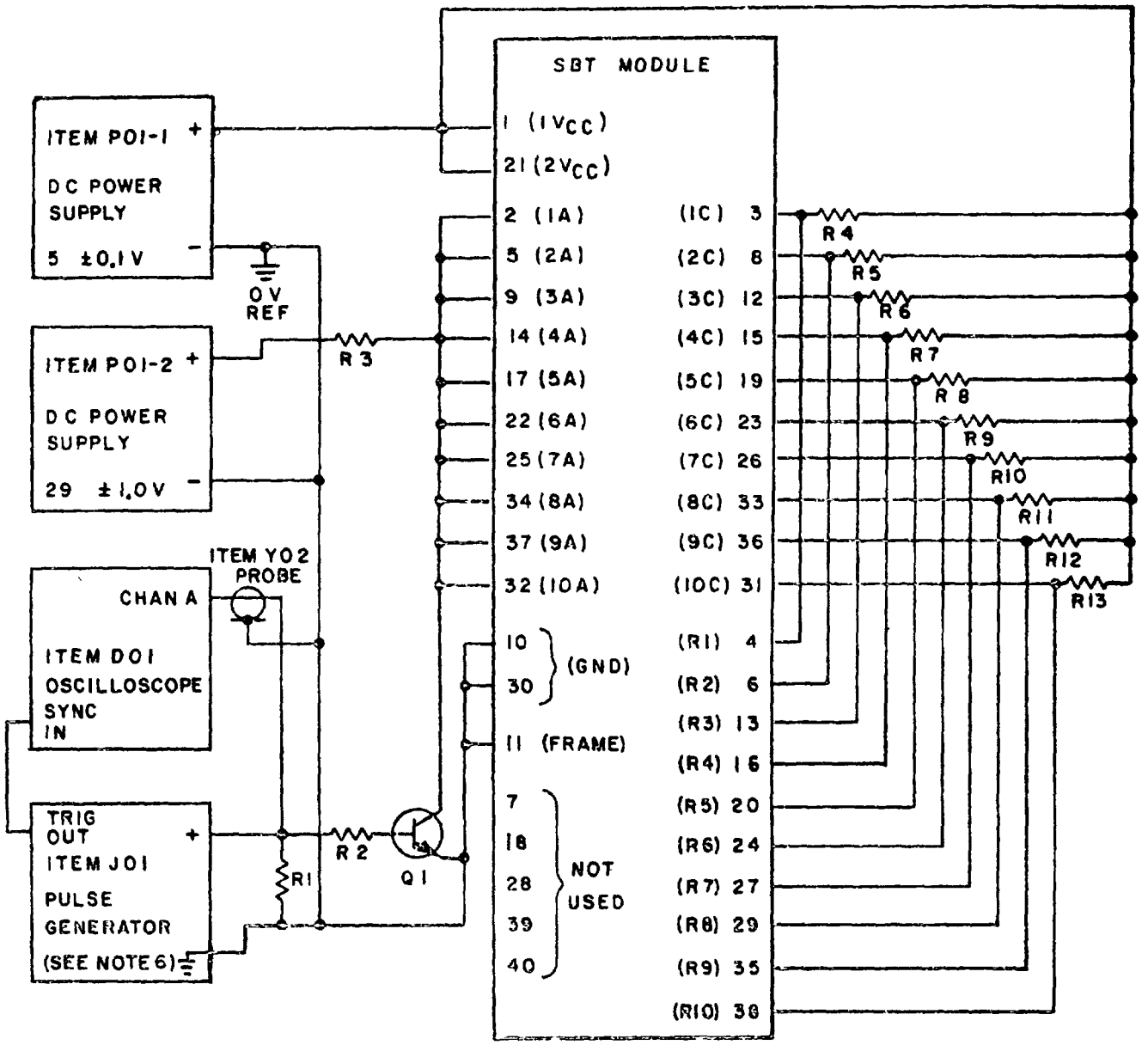
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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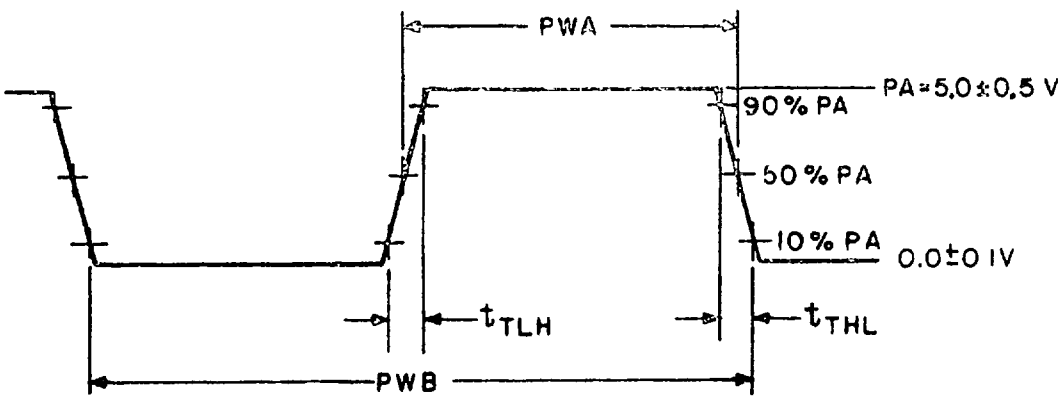
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NOTES:
1. R1 = 50Ω ±5%, 1 W, noninductive.
2. R2 = 511Ω ±1%, 1/4 W.
3. R3 = 270Ω ±5%, 5 W.
4. R4 - R13 = 383Ω ±1%, 1/4 W.
5. Q1 = 2N2364 or equivalent.
6. Adjust item J01 output to obtain the waveform shown on figure 3.

* FIGURE 2. Operating temperature, component temperature and life test circuit.

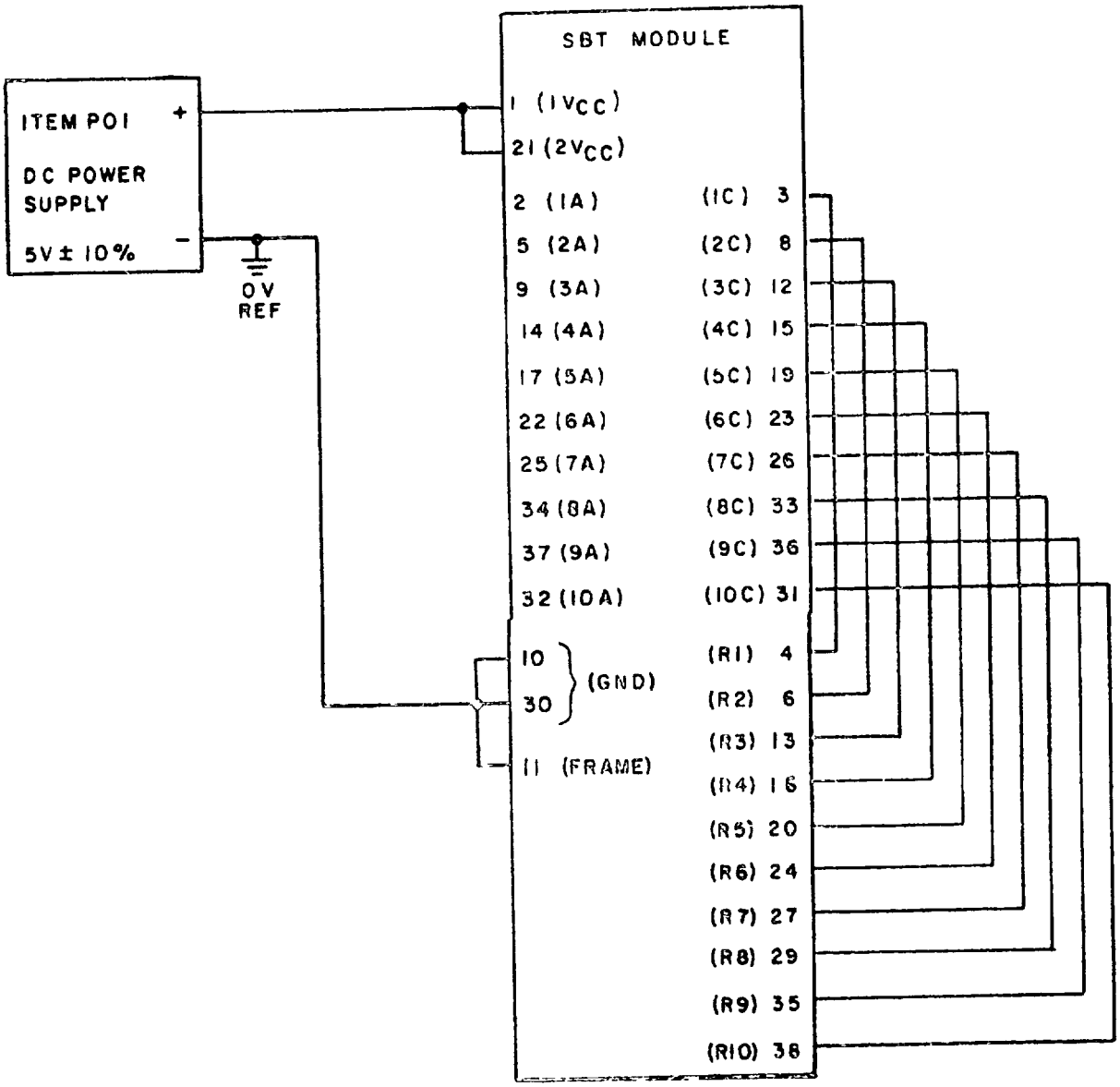
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NOTE: Input pulse parameters:
PWA = 1 ±0.1 ms.
PWB = 2 ±0.2 ms.
$t_{TLH} = t_{THL}$ = 100 ±10 ns.

* FIGURE 3. Waveform for figure 2 and tables VIII through X.

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* FIGURE 4. Vibration test circuit.

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* TABLE II. Breakdown voltage check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1C Breakdown voltage check	3	0	5	µA	5
A2	2C	8				
A3	3C	12				
A4	4C	15				
A5	5C	19				
A6	6C	23				
A7	7C	26				
A8	8C	33				
A9	9C	36				
A10	10C	31				

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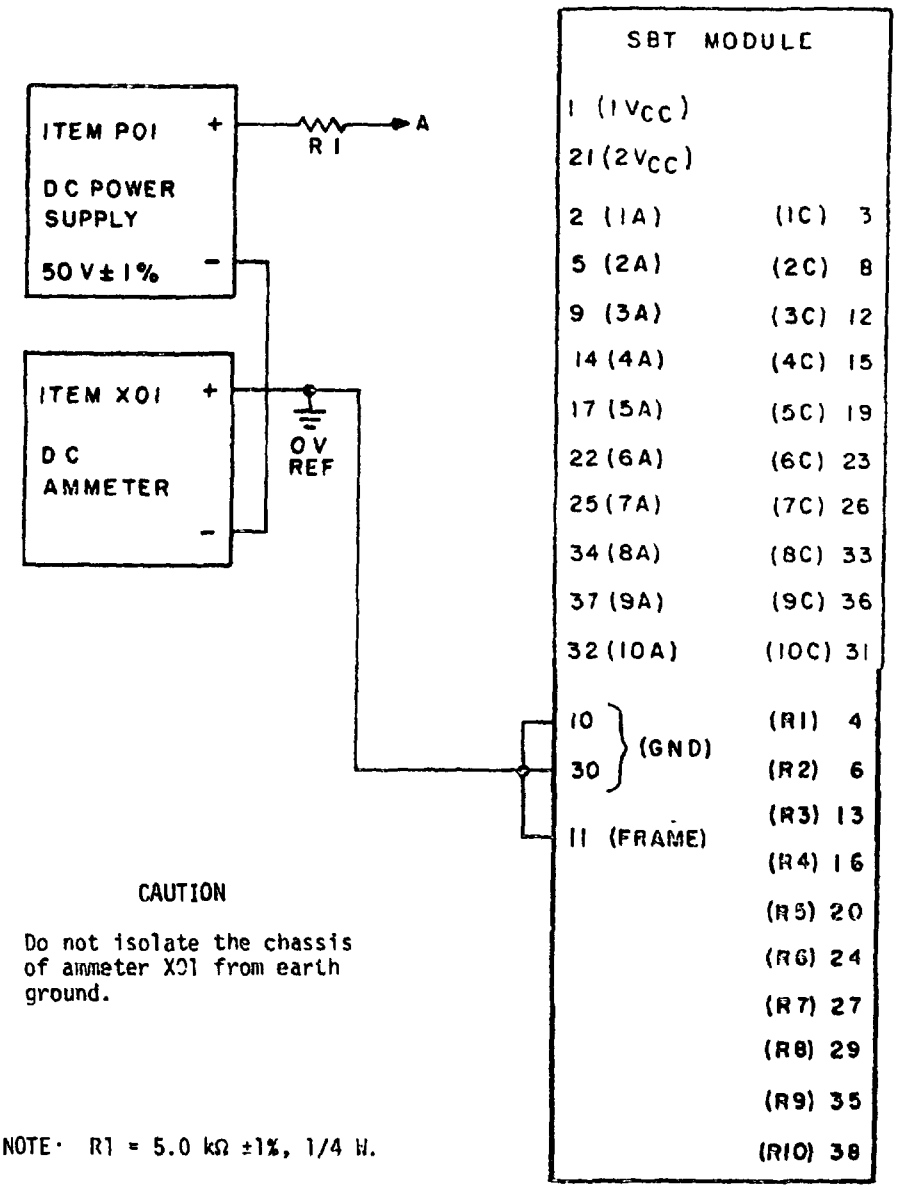


FIGURE 5. Breakdown voltage check test circuit for table II.

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* TABLE III. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	1C Output voltage, low level (V_{IN} = 22 V, I_C = 50 mA)	3,4,6	2	Open	0	250	mV	6
B2	2C	4,6,8	5					
B3	3C	12,13,16	9					
B4	4C	13,15,16	14					
B5	5C	19,20,24	17					
B6	6C	20,23,24	22					
B7	7C	26,27,29	25					
B8	8C	27,29,31	32					
B9	9C	33,35,38	34					
B10	10C	35,36,38	37					

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* TABLE IV. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	1C Output voltage, high level (V_{IN} = Open)	3,4	Open	Open	4.90	5.05	V	6
C2	2C	6,8	↓	↓	↓	↓	↓	↓
C3	3C	12,13						
C4	4C	15,16						
C5	5C	19,20						
C6	6C	23,24						
C7	7C	26,27						
C8	8C	29,31						
C9	9C	33,35						
C10	10C	36,38						

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TABLE V. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1C Output voltage, high level (V_{IN} = 6 V)	3,4	2	Open	4.90	5.05	V	6
D2	2C	6,8	5	↓	↓	↓	↓	↓
D3	3C	12,13	9					
D4	4C	15,16	14					
D5	5C	19,20	17					
D6	6C	23,24	22					
D7	7C	26,27	25					
D8	8C	29,31	32					
D9	9C	33,35	34					
D10	10C	36,38	37					

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* TABLE VI. Input current (low-level) test.

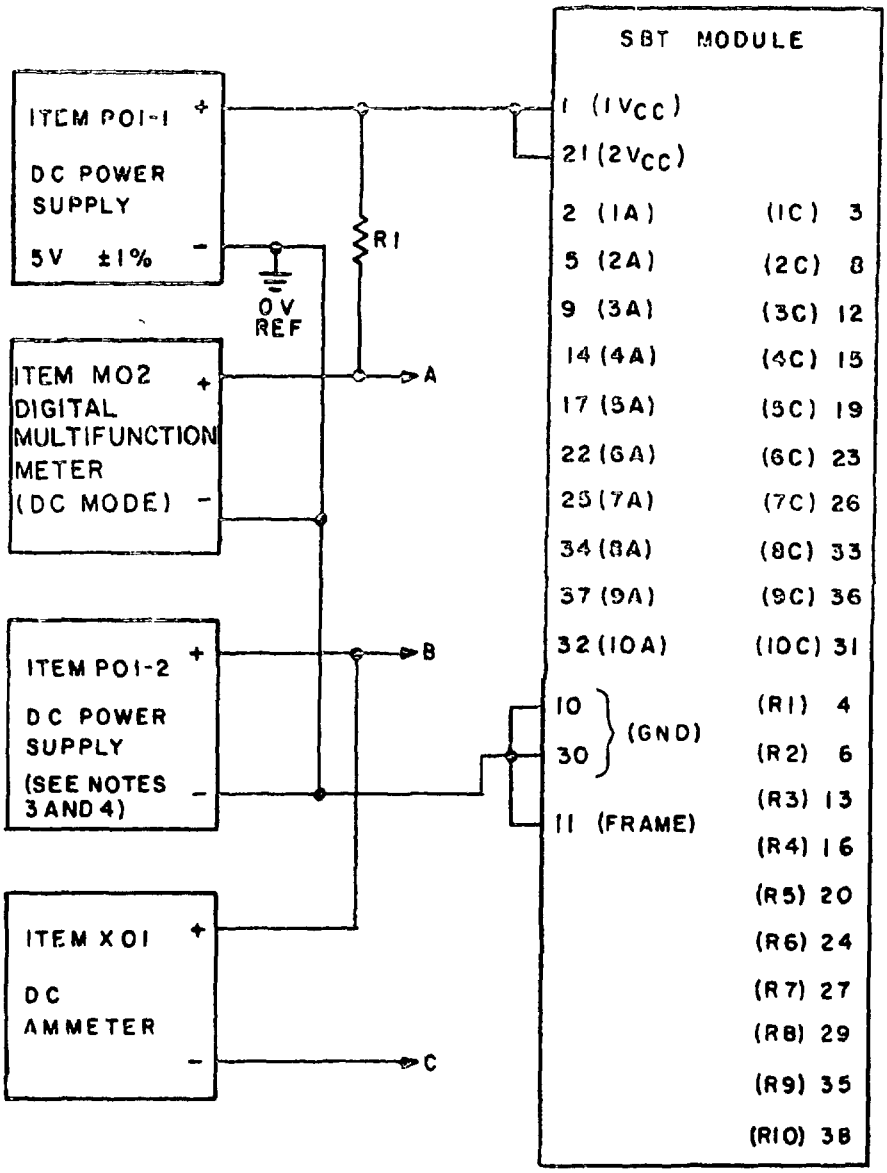
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1A	Input current, low level (V_{IN} = 6 V)	3,4	Open	2	0	285	µA	6
E2	2A		6,8		5				
E3	3A		12,13		9				
E4	4A		15,16		14				
E5	5A		19,20		17				
E6	6A		23,24		22				
E7	7A		26,27		25				
E8	8A		29,31		32				
E9	9A		33,35		34				
E10	10A		36,38		37				

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* TABLE VII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, high level (V_{IN} = 22 V)	3,4,6	Open	2	0.10	1.15	mA	6
F2	2A	4,6,8		5				
F3	3A	12,13,16		9				
F4	4A	13,15,16		14				
F5	5A	19,20,24		17				
F6	6A	20,23,24		22				
F7	7A	26,27,29		25				
F8	8A	27,29,31		32				
F9	9A	33,35,38		34				
F10	10A	35,36,38		37				

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NOTES:
1. R1 = 383Ω ±1%, 1/4 W, for test conditions C, D, and E.
2. R1 = 100Ω ±1%, 1/4 W, for test conditions B and F.
3. For test conditions D and E, adjust item PO1-2 to 6 V ±1%.
4. For test conditions B and F, adjust item PO1-2 to 22 V ±1%.

*FIGURE 6. Output voltage (low-level and high-level) and input current (low-level and high-level) test circuit for tables III through VII.

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* TABLE VIII. Input pulse noise suppression test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	1A Input pulse noise suppression 1/	2	3,4	4.75	5.05	V	7
G2	2A	5	6,8				
G3	3A	9	12,13				
G4	4A	14	15,16				
G5	5A	17	19,20				
G6	6A	22	23,24				
G7	7A	25	26,27				
G8	8A	32	29,31				
G9	9A	34	33,35				
G10	10A	37	36,38				

1/ Measure the minimum output voltage level at test point B.

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* TABLE IX. Propagation delay time (t_{PHL}) test.

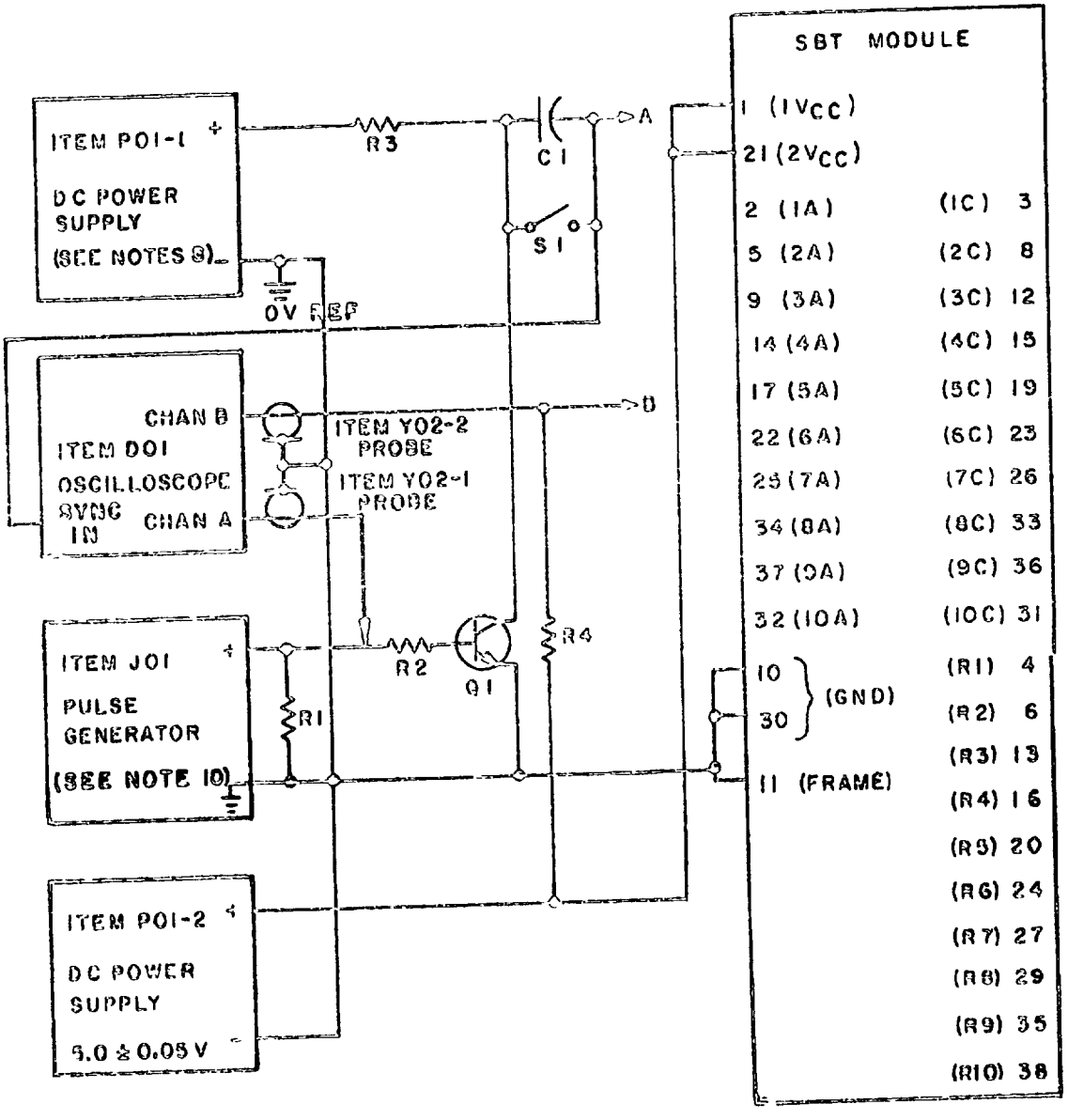
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Propagation delay time (t_{PHL})	2	3,4	250	450	µs	7,8
H2	↓	5	6,8	↓	↓	↓	↓
H3		9	12,13				
H4		14	15,16				
H5		17	19,20				
H6		22	23,24				
H7		25	26,27				
H8		32	29,31				
H9		34	33,35				
H10		37	36,38				

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* TABLE X. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Propagation delay time (t_{PLH})	2	3,4	96	172	µs	7,8
J2	↓	5	6,8	↓	↓	↓	↓
J3		9	12,13				
J4		14	15,16				
J5		17	19,20				
J6		22	23,24				
J7		25	26,27				
J8		32	29,31				
J9		34	33,35				
J10		37	36,38				

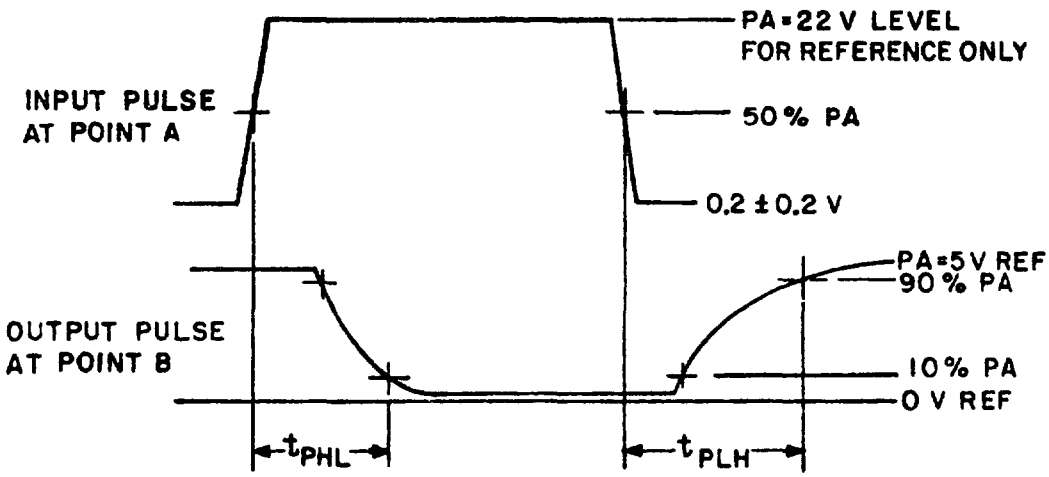
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 511Ω ±1%, 1/4 W.
3. R3 = 270Ω ±5%, 5 W.
4. R4 = 383Ω ±1%, 1/4 W.
5. C1 = 3900 pF ±10%, 100 Vdc.
6. Q1 = 2N2364 or equivalent.
7. S1 = SPST switch.
8. Adjust item P01-1 as follows:
   32 V ±1% for test conditions G1-G10.
   22 V ±1% for test conditions H1-H10 and J1-J10.
9. Set switch S1 to open position for test conditions G1-G10 and closed for test conditions H1-H10 and J1-J10.
10. Adjust item J01 output to obtain the waveform shown on figure 3.

* FIGURE 7 Input pulse noise suppression and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables VIII, IX, and X.

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NOTES:
1. After adjusting the input pulse, connect item Y02-1 to test point A.
2. Simultaneously observe the module input (point A) and output (point B) and measure the propagation delay time.

* FIGURE 8. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveform for tables IX and X.

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* TABLE XI. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
K1	Isolation	10,30	11	10	∞	MΩ	9
K2	↓	7,18,28,39, and 40	All other pins	↓	↓	↓	↓
K3		2,3,5,8,9,10,12,14, 15,17,19,22,23,25, 26,30,31,32,33,34, 36, and 37	1,4,6,13,16,20,21, 24,27,29,35, and 38				
K4		2,3,5,8,9,10,12,14, 15,17,19	22,23,25,26,30,31, 32,33,34,36,37				
K5		1,4,6,13,16,20	21,24,27,29,35,38				
K6	↓	11	All other pins	↓	↓	↓	↓

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TABLE XII. Resistance test.

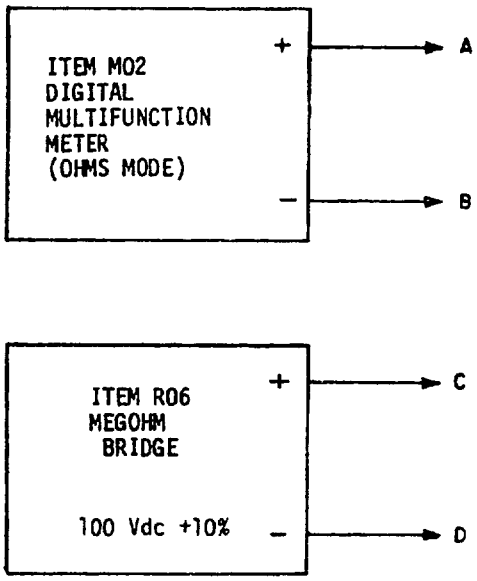
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	R1	Resistance	1	4	4.588	4.692	kΩ	9
L2	R2	↓	1	6	↓	↓	↓	↓
L3	R3		1	13				
L4	R4		1	16				
L5	R5		1	20				
L6	R6		21	24				
L7	R7		21	27				
L8	R8		21	29				
L9	R9		21	35				
L10	R10		21	38				

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TABLE XIII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M	Continuity	11	Frame	0	1	Ω	9

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SBT MODULE	
1 ($1V_{CC}$)	
21 ($2V_{CC}$)	
2 (1A)	(1C) 3
5 (2A)	(2C) 8
9 (3A)	(3C) 12
14 (4A)	(4C) 15
17 (5A)	(5C) 19
22 (6A)	(6C) 23
25 (7A)	(7C) 26
34 (8A)	(8C) 33
37 (9A)	(9C) 36
32 (10A)	(10C) 31
10, 30 (GND)	(R1) 4
	(R2) 6
11 (FRAME)	(R3) 13
	(R4) 16
	(R5) 20
	(R6) 24
	(R7) 27
	(R8) 29
	(R9) 35
	(R10) 38

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition M, penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative terminal.

FIGURE 9. Isolation, resistance, and continuity test circuit for tables XI, XII, and XIII.

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